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D. JAYASREE

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S.NO

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16/7/26

```
C:\Users\DELL>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49675        HDC012H113:49676       ESTABLISHED
  TCP    127.0.0.1:49676        HDC012H113:49675       ESTABLISHED
  TCP    127.0.0.1:49677        HDC012H113:49678       ESTABLISHED
  TCP    127.0.0.1:49678        HDC012H113:49677       ESTABLISHED
  TCP    127.0.0.1:49682        HDC012H113:49683       ESTABLISHED
  TCP    127.0.0.1:49683        HDC012H113:49682       TIME_WAIT
  TCP    127.0.0.1:63905        HDC012H113:63906       ESTABLISHED
  TCP    172.16.75.53:1521      HDC012H113:57609       TIME_WAIT
  TCP    172.16.75.53:7680      172.16.75.35:49193     TIME_WAIT
  TCP    172.16.75.53:7680      172.16.75.85:60908     ESTABLISHED
  TCP    172.16.75.53:7680      172.16.75.105:58668    TIME_WAIT
  TCP    172.16.75.53:7680      172.16.77.24:55750     TIME_WAIT
  TCP    172.16.75.53:7680      172.16.77.33:56991     TIME_WAIT
  TCP    172.16.75.53:7680      172.16.77.34:49585     TIME_WAIT
  TCP    172.16.75.53:7680      172.16.77.37:59924     TIME_WAIT
  TCP    172.16.75.53:7680      THALAPATHY-VIJA:61829  ESTABLISHED
  TCP    172.16.75.53:49414     a-0213:25.242:https    ESTABLISHED
  TCP    172.16.75.53:49960     pnmaaa-bf-in-f10:https ESTABLISHED
  TCP    172.16.75.53:50000     sa-in-f188:5228        ESTABLISHED
  TCP    172.16.75.53:56658     172.16.75.115:ms-do    ESTABLISHED
  TCP    172.16.75.53:57689     HDC012H113:1521        ESTABLISHED
  TCP    172.16.75.53:59971     172.16.75.115:ms-do    ESTABLISHED
  TCP    172.16.75.53:60774     a-150-223-97:https     ESTABLISHED
  TCP    172.16.75.53:60779     172.16.75.115:ms-do    ESTABLISHED
  TCP    172.16.75.53:60781     172.16.75.116:ms-do    TIME_WAIT
  TCP    172.16.75.53:63934     lcmaaa-an-in-f3:https  ESTABLISHED
  TCP    172.16.75.53:63947     150.171.110.83:https   ESTABLISHED
  TCP    172.16.75.53:63949     172.16.75.105:ms-do    ESTABLISHED
  TCP    172.16.75.53:63951     a23-211-105-64:https   ESTABLISHED
  TCP    172.16.75.53:63952     a23-211-105-64:https   ESTABLISHED
  TCP    172.16.75.53:63958     52.123.128.14:https    ESTABLISHED
  TCP    172.16.75.53:63960     20.42.73.30:https      ESTABLISHED
  TCP    172.16.75.53:63961     20.42.73.30:https      ESTABLISHED
  TCP    172.16.75.53:63962     52.123.129.14:https    TIME_WAIT
  TCP    172.16.75.53:63963     172.16.75.116:ms-do    SYN_SENT
  TCP    172.16.75.53:63965     172.16.75.99:ms-do     CLOSE_WAIT
  TCP    172.16.75.53:64969     dns:https              ESTABLISHED
  TCP    172.16.75.53:65244     172.16.75.216:ms-do    ESTABLISHED
  TCP    172.16.75.53:65246     172.16.77.70:ms-do     ESTABLISHED
```

```
C:\Users\DELL>nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

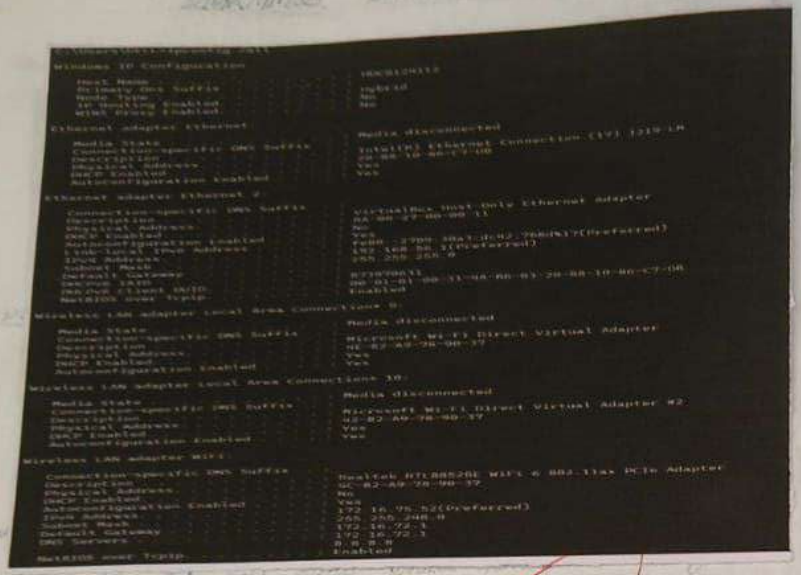
  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

```
C:\Users\DELL>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.
```



```
Windows PowerShell
indows PowerShell
opyright (C) Microsoft Corporation. All rights reserved.

nstall the latest PowerShell for new features and improves

S C:\Users\DELL> arp -a

nterface: 192.168.56.1 --- 0x11
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.0.253           01-00-5e-00-00-fd     static
  230.0.0.1             01-00-5e-00-00-01     static
  239.0.0.8             01-00-5e-00-00-08     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.67.250        01-00-5e-7f-43-fa     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

nterface: 172.16.75.48 --- 0x13
  Internet Address      Physical Address      Type
  172.16.72.1           7c-5a-1c-cf-be-41     dynamic
  172.16.79.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  ---.---.---.---       --  --  --

PS C:\Users\DELL> hostname
HDC0124108
PS C:\Users\DELL>
```

5) **Netstat:**

Netstat displays a variety of statistics about a computer TCP/IP connection. It is a command line tool for monitoring network connection incoming and outgoing as well as viewing routing tables.

eg. netstat - r

6) **nslookup:**

It is a tool used to perform DNS lookups and display DNS details, such as the IP address of a particular computer. nslookup can operate in two modes: interactive and non-interactive.

7) **Pathping:**

Pathping is unique to windows, and is basically a combination of Tracer commands. Pathping traces the route to the destination address then the second test of each routes along the way, gathering statistics on the rate of data hop.

8) **Ping:**

It is the command to test connection nodes. Ping use ICMP to communicate.

1) #Ping hostname (Ping localhost)

2)# Ping ip addess (Ping 4.2.2.2)

3)# Ping fully qualified domain name (ping www.facebook.com)

9) **Route:**

Route command is used to show/manipulate the IP routing table. It is to setup static routes to specific host or networks via an interface.

10) **IP:**

The ip command is one of the basic commands every administrator will need in daily work, from setting up new systems and assigning IPs to trouble shooting existing system.

Ex No: 1
16/7/26

BASIC NETWORKING COMMANDS IN LINUX AND WINDOWS OPERATING SYSTEMS

AIM:

To study and execute basic networking commands in Linux for testing network connectivity.

1) arp -a:

ARP is short form of address resolution protocol, It will show the IP address of the computer along with the IP address and MAC address of your router.

2) hostname:

This is the simplest of all TCP/IP commands. It simply displays the host of the computer.

3) ipconfig /all:

This command display detailed configuration information about the connection including routes, gateway, DNS, DHCP, and type of ethernet of system.

4) nbtstat -a:

This command helps solve problems with NetBIOS name resolution.

```
                statistics.

PS C:\Users\DELL> nslookup
Default Server:  UnKnown
Address:  172.16.72.1
```

```
> ping
Server:  UnKnown
Address:  172.16.72.1
```

```
                statistics.

PS C:\Users\DELL> nslookup
Default Server:  UnKnown
Address:  172.16.72.1
```

```
C:\Users\DELL>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

```
Command Prompt
239.255.255.250       01-00-5e-7f-ff-fa     static

C:\Users\DELL>hostname
```

(1) **Ifconfig:**

The ifconfig command was in a staple in many sysadmin's tool kit for configuring and trouble shooting networks. It has since been replaced by the ip commands discussed above.

(2) **mtr:**

MTR is a program with a command-line interface that serves as a network diagnostic and trouble shooting tool. This command combines the functionality of the Ping and traceroute commands.

(3) **tcpdump**

the tcpdump command is designed for capturing and displaying packets.

Result:

The basic command for Linux and window are sucessfully completed.

EXP NO: 2
16/7/26

STUDY OF DIFFERENT TYPES OF NETWORK CABLES

AIM:

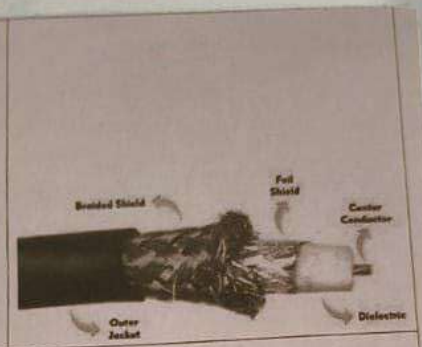
To study and understand different types of network cable, to make your own ethernet cross-over cable / straight cable.

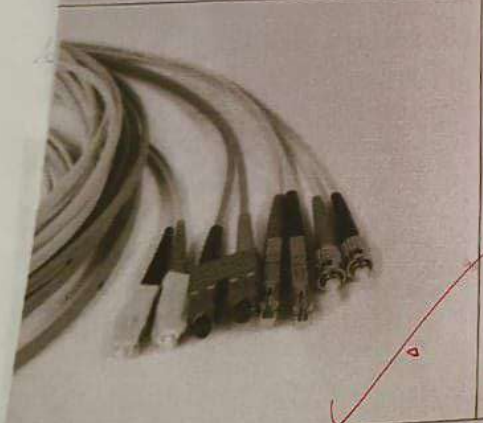
a) understand different type of network cable

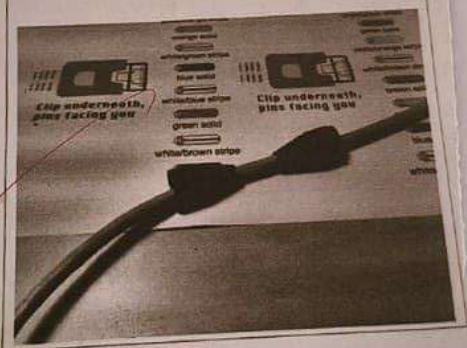
Different type of cables used in networking are.

1) Unshielded twisted pair (UTP) cable.
2) Shielded twisted pair (STP) cable.
3) Coaxial cable
4) Fiber optic cable.

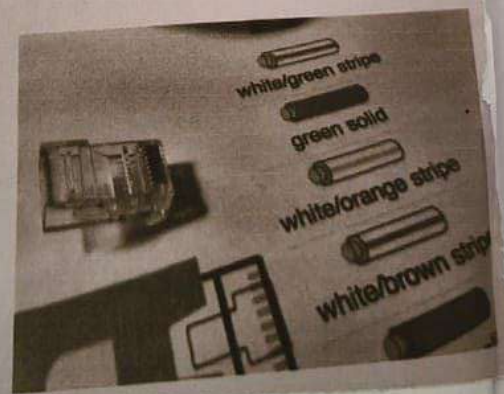




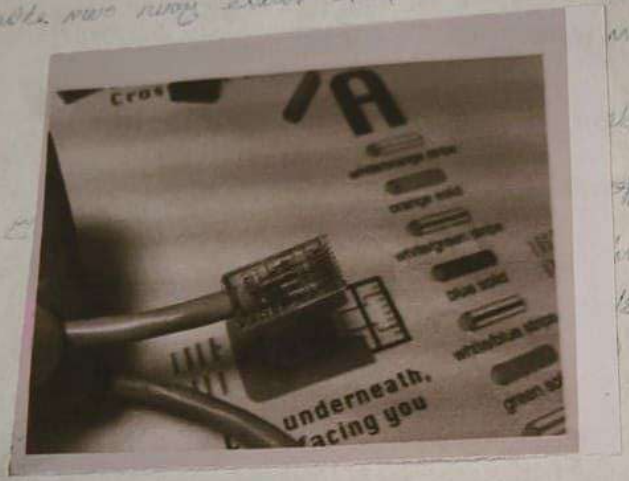


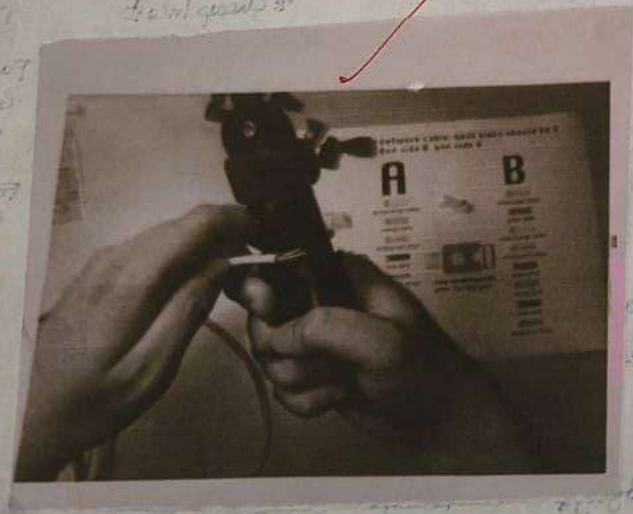






Cable type	category	Maximum Data Transmission	Adv/Dis	App/Use
UTP	Category 3	10bps	Adv * cheap in cost * easy to install Disadv * More prone to (EMI) * Noise	10Base-T Ethernet
	Category 5	up to 1000Mbps		Fast ethernet, Gigabit ethernet.
	Category 5e	1 Gbps		Fast ethernet, Gigabit Ethernet.
SFP	category 6,6a	10Gbps	Adv * Shielded * Faster than UTP	Gigabit Ethernet, 10G Ethernet
SSTP	Category 7	10Gbps	Disadv * Expensive * Greater	Gigabit Ethernet, 10G Ethernet
Fibre optics Cable	single mode multi mode	100Gbps	Adv * High speed Disadv * Expensive	Maximum distance fibre cable





Procedure:

1. To start construction of the device, begin by threading shields onto the cable
2. Next strip approximately 1.5cm of cable shielding from both ends. The crimping tool has a second area to complete this task.
3. After, you will need to untangle the wires, there should be four twisted pairs. Referencing back to the sheet, arrange them from top to bottom. one end should be in arrangement A and the other in B.
4. Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing them into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you, as shown.
5. Next, push the cable right in the notch at the end of the plug needs to be just over the cable shielding, and if it isn't, that means that you stripped off too much shielding. Simply snip the cables back a little more.

EXNO: 3b2
Date: 23/8/25

Switch

AIM:

To study the behaviour of a switch by observing how it forwards data packets using cisco packet tracer.

STEPS:

i) Open cisco packet tracer.

ii) Drag and place 1 switch from 4 Generic PCs into the workspace.

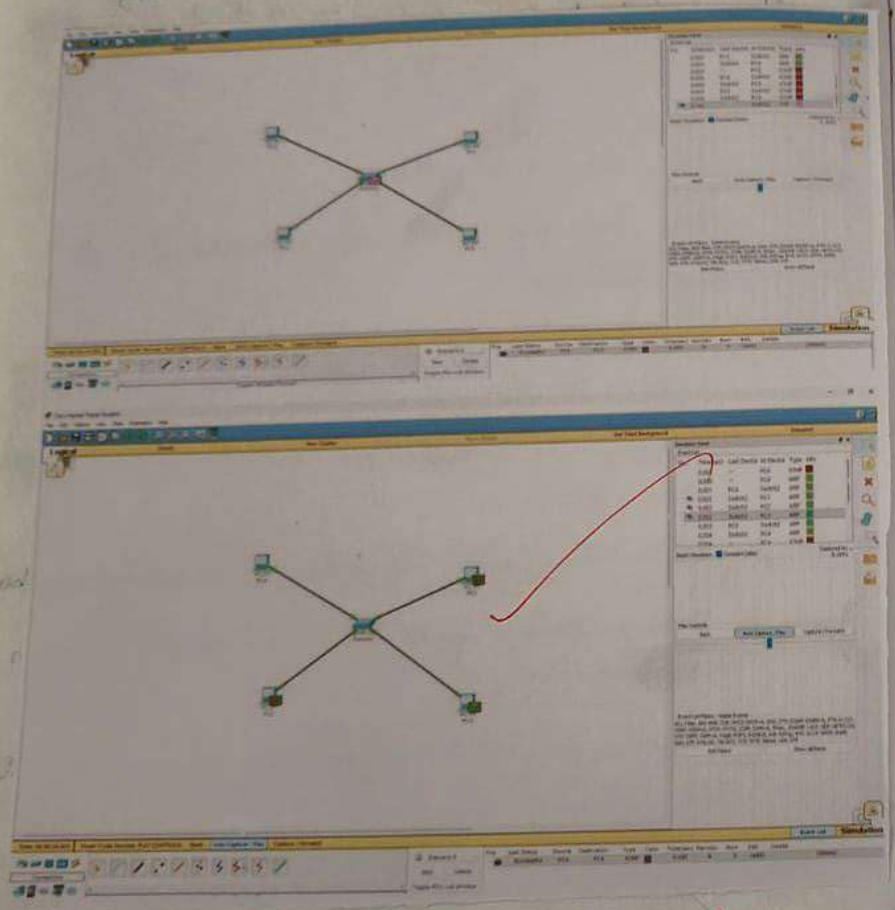
iii) Connect all PCs to the switch using copper straight - through cables.

iv) Configure IP addresses for all PCs in the same network.

v) Select Add simple PDU from the common tools bar.

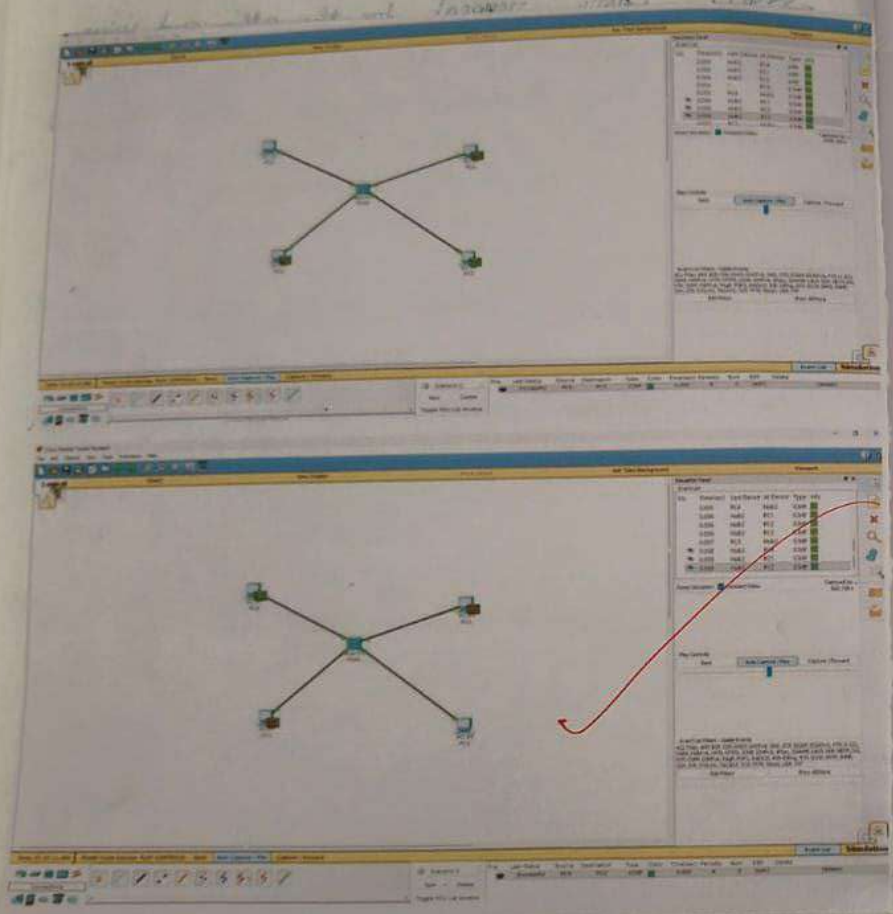
vi) Send a packet from one PC (source) to another PC (destination)

vii) Switch to simulation mode and observe how the switch forwards the packet.



RESULT:

The switch forwards the packet only to the destination PC by using its MAC address table. This reduces unnecessary network traffic and improves communication efficiency.



EXNO: 3b1
23/7/26

HUB

AIM:

To study the behaviour of a hub by observing how it forwards data packets using Cisco packet tracer.

STEPS:

i) Open cisco packet tracer.

ii) Drag and place 1 hub and 4 Generic PCs into the workspace.

iii) Connect all PCs to the hub using copper straight-through cables.

iv) Configure IP addresses for all PCs in the same network.

v) Select add simple PDU from the common tools bar.

vi) Send a packet from one PC (source) to another PC (destination)

vii) Switch to simulation mode and observe how the hub forwards the packet.

RESULT:

The hub broadcasts the received packet to all connected devices. Every PC receives the packet, but only the destination PC accepts while the others discard it.

Step 6 After the wires are accurately sitting inside the plug, insert it into the crimping tool and push down.

Step 7: Lastly, repeat for the other end using diagram B (to make a cross over cables) / using diagram A (to make a straight through cable). To test it, plug it in and attempt to connect two devices directly.

RESULT:

Thus, different types of network cables were studied successfully, and straight through and cross over ethernet cables were prepared and tested successfully.

Ex No: 3b3
23/7/26

HUB & SWITCH (comparison)

AIM:

To compare the packet forwarding behaviour of a hub and a switch using cisco packet tracer.

Steps:

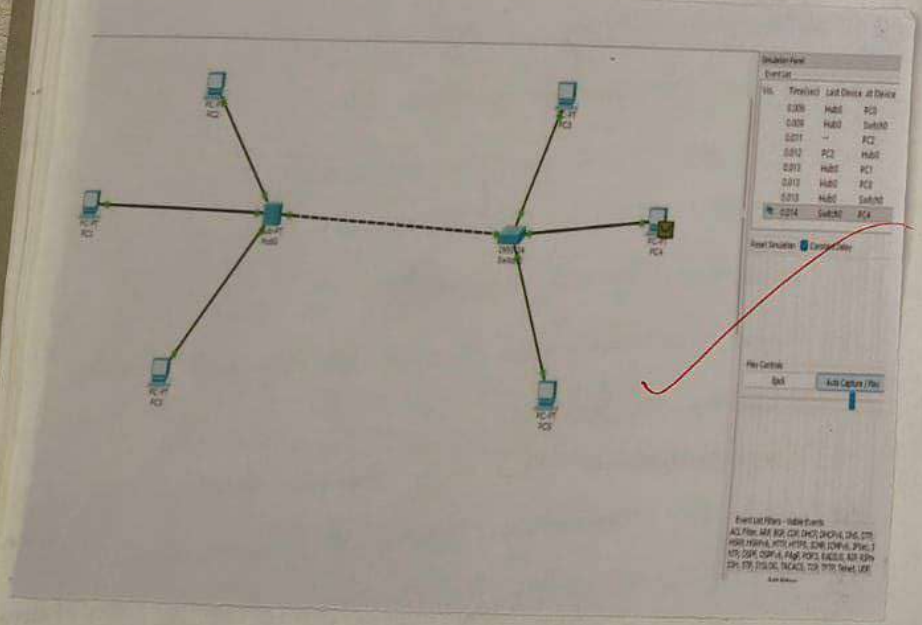
i) Create one network using a hub and another using a switch, each connected to 4 PCs.

ii) Configure IP addresses for all PCs in both networks.

iii) Send a simple PDU between two PCs in each network.

iv) Observe packet transmission in simulation mode.

v) Compare how the hub and switch forward packets.



RESULT:

The hub broadcasts packets to all connected devices, whereas the switch forwards packets only to the intended destination.

Ex:3A To Understand the Environment of cisco packet tracer design a single network.

AIM:

To understand the Cisco Packet Tracer environment and design a simple Computer network by connecting network devices, configuring IP address, and verifying network communication

ALGORITHM:

1) Start Cisco Packet Tracer.
2) Place the required network devices in the workspace.
3) Connect the devices using appropriate network cables.
4) Configure IP address and subnet mask for all devices
5) Configure the router interfaces.
6) Save the network topology.
7) Test the connectivity using the Ping Command.
8) Observe packet transmission in simulation Mode.
9) Verify successful communication between all connected devices.
10) End the experiment.

Procedure:

1) open cisco Packet Tracer.
2) Select two PCs and one switch from the device panel.
3) Drag and place the devices in the workspace.
4) Connect each PC to the switch using copper Straight - Through cables.
5) Assign IP address to the PCs
   - PC0: 192.168.1.1 / 255.255.255.0
   - PC1: 192.168.1.2 / 255.255.255.0
6) Save the network
7) Open the command Prompt on PC0 and execute.
   Ping 192.168.1.2
8) Check whether the ping receives successful replies.
9) Switch to simulation Mode to observe packet flow.

OUTPUT:

- The network topology was created successfully
- IP address were assigned correctly to the PCs
- the ping command returned reply from 192.168.1.2, indicating succesful communication.
- Simulation mode displayed succesful packet transmission between the devices.

RESULT: The cisco Packet Tracer environment was successfully understood, and a simple network was designed, configured, and tested. the devices communicated succesfully.

EXPNO:4
30/7/26

CONFIGURE A LAN USING SWITCH AND ETHERNET CABLES

AIM:

Setup and configure a LAN (Local Area Network) using a switch and ethernet cables in your lab.

What is a LAN?

A Local Area Network (LAN) refers to a network that connects devices within a limited area such as an office building, school or home. It enables users to share resources, including data, printers, and internet access. LAN connects devices to promote collaboration and transfer information between users, such as computers, printers, servers and switches. A local area network (LAN) switch serves as the primary connecting device, managing and directing communication within the local network. Each connected device on a LAN switch can communicate directly with each other, allowing for fast and secure data transfer.

STEPS TO SET UP A LAN:

Step:1

Plan and design an appropriate network topology taking into account network requirement and equipment location.

Step:2

Take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for networks of these sizes, and 4 ethernet cables.

Step: 3

connect the computers to network switch via an ethernet cable, which is as simple as plugging one end of the ethernet cable into the computer and the other end into the network switch.

Step: 4

Assign IP address to your PCs

1) Log on to the client computer as administrator or as owner.
2) click network and internet connections.
3) Right click local area connection/ ethernet → Go to properties → select use the following ip address option and assign ip address.
Similarly assign IP address to all the PCs connected to switch.

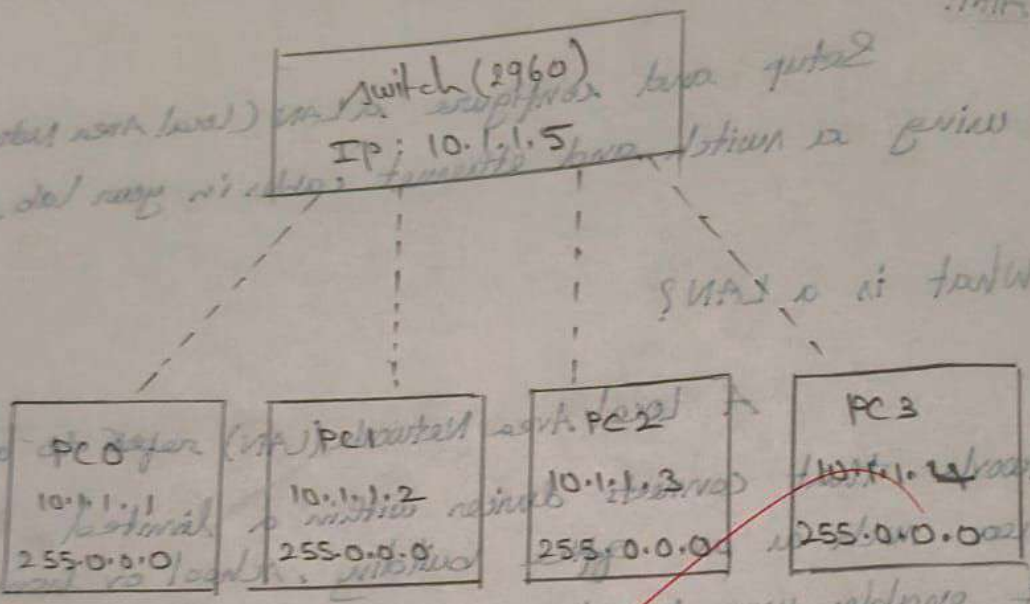
PC1 - IP address : 10.1.1.1, subnet mask 255.0.0.0
PC2 - IP address : 10.1.1.2, subnet mask 255.0.0.0
PC3 - IP address : 10.1.1.3, subnet mask 255.0.0.0
PC4 - IP address : 10.1.1.4, subnet mask 255.0.0.0

Step: 5

Configure a network switch.

Step: 6

Check the connectivity between switch and other machine by using ping command in the command prompt of the device.


Switch (2960)
Ip: 10.1.1.5
PC 0
10.1.1.1
255.0.0.0
PC 1
10.1.1.2
255.0.0.0
PC 2
10.1.1.3
255.0.0.0
PC 3
10.1.1.4
255.0.0.0

Ex No: 5
30/7/26

Experiments on Packet capture tool : Wireshark.

AIM: Experiments on Packet capture tool : wireshark.

Packet Sniffer:

- Sniffs messages being sent/received from/ by your computer.
- Store and display the contents of the various protocol fields in the messages
- Passive program.
  - never sends packet itself.
  - no packets addressed to it.
  - receives a copy of all packets

Packet Sniffer structure Diagostic Tools

- Tcpdump
- wireshark

Description:

wireshark: wireshark, a network analysis tool formerly known as ethereal, captures packets in real time and display them in human-readable format. wireshark includes filters, color coding, and other features that let you dig deep into network traffic and inspect individual packets.

What we can do with wireshark:

→ Capture network traffic.
→ Decode packet protocols using dissectors.
→ Watch smart statistics.

Exercise no:)
3/9/24
c1) Interconnectivity with routers in CISCO PACKET TRACER simulator.

Design and configure a simple Internetwork using a router.

In this network, a router and 2 PCs are used. Computers are connected with routers using a copper straight-through cable. After forming the network, to check network connectivity a simple PDU is transferred from PC0 to PC1.

Procedure:

Step:1 (Configuring router: 1):

1.) Select the router and open CLI.
2) Press ENTER to start configuring router.
3) Type enable to activate the privileged mode.

Router1 command line interface:

Router>enable
Router# config t
Enter configuration commands, one per line.

Step-2 (Configuring PCs):

1. Assign IP addresses to every PC in the network.
2. Select the PC, Go to the desktop and select IP configuration and assign an IP address, Default gateway, subnet mask.
3. Assign the default gateway of PC0 as 192.168.10.1
4. Assign the default gateway of PC1 as 192.162.20.1

Step:3 (Connecting PCs with router):

1. Connect FastEthernet0 port of PC0 with Fast Ethernet 0/0 port of router1 using a copper straight through cable.

Router Configuration Table:

Devicename	IP address FastEthernet 0/0	Subnet Mask	IP address FastEthernet 0/1	Subnet Mask
Router1	192.168.10.1	255.255.255.0	192.168.20.1	255.255.255.0

Experiment 3
3/9/24

Internetworking with routers in CISCO PACKET TRACER simulator.

a) Design and configure a simple internetwork using a router.

In this network, a router and 2 PCs are used. Computers are connected with routers using a copper straight-through cable. After forming the network, to check network connectivity a simple PDU is transferred from PC0 to PC1.

Procedure:

Step:1 (Configuring router 1):

1.) Select the router and open CLI.
2) Press ENTER to start configuring router1
3) Type enable to activate the privileged mode.

Router1 command line interface:

Router>enable
Router# config t
Enter configuration commands, one per line.

Step-2 (Configuring PCs):

1. Assign IP addresses to every PC in the network.
2. Select the PC, Go to the desktop and select IP configuration and assign an IP address, Default gateway, subnet mask.
3. Assign the default gateway of PC0 as 192.168.10.1
4. Assign the default gateway of PC1 as 192.162.20.1

Step:3 (Connecting PCs with router):

1. connect FastEthernet0 port of PC0 with Fast Ethernet 0/0 port of router1 using a copper straight through cable.

Router Configuration Table:

Devicename	IP address FastEthernet 0/0	subnet Mask	IP address FastEthernet 0/1	subnet Mask
Router1	192.168.10.1	255.255.255.0	192.168.20.1	255.255.255.0

PC configuration table:

Device name	IP address	Subnet mask	Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1

Designed network topology:

Simulation of designed network topology.

Result:

Thus, the interoperability with routers in CISCO PACKET TRACER simulator has been executed successfully.

```
# no shutdown
# exit
interface FastEthernet 0/1
ip address [IP address] [subnet mask]
no shutdown
exit
```

Testing the Network:

Now that our network topology is configured, we can test the network. Open a command prompt on each PC and try to ping the other PC. If the ping is successful, then the network is functioning properly.

Result:

Thus the implementation of subnetting in CISCO Packet tracer simulator has been verified

OUTPUT:

Wireless network : mothiranetwork

Security : WEB

WEB : 0123456789

PC0 connected successfully

PC1 connected successfully

PC2 connected successfully

PC0 > ping 192.168.0.3

Reply from 192.168.0.3

Result:

SUBNETTING in CISCO PACKET TRACER

Aim:

To implementation of subnetting in cisco packet tracer simulator.

Creating a network topology:

The first step in implementing classful subnetting is to create a network topology in Packet tracer.

Adding the Devices:

Once we have created our network topology, we can add devices to it. Here, we will be adding routers, switches, and PC's. To add a device, select the device from the bottom left corner and drag it onto the network topology. Then, connect the devices by choosing a cable from one device's port to another device's port.

Subnetting:

To subnet the network address of 192.168.1.0/24 to provide enough space for at least 5 addresses for end devices, the switch, and the router, we can use a /27 subnet mask.

The IP addressing for the network shown in the topology can be as follows:

- Router R1:
- GigabitEthernet0/0: 192.168.1.1
- Gigabit Ethernet0/1: 192.168.2.1

- Switch S1:
- FastEthernet 0/1 : 192.168.1.0/27
- PC1 : 192.168.1.11
- PC2 : 192.168.1.12
- PC3 : 192.168.1.13
- PC4 : 192.168.1.14
- PC5 : 192.168.1.15
- FastEthernet 0/2 : 192.168.2.0/27
- PC1 : 192.168.2.11
- PC2 : 192.168.2.12
- PC3 : 192.168.2.13
- PC4 : 192.168.2.14
- PC5 : 192.168.2.15
- Router R2 :
- FastEthernet 0/0 : 192.168.3.1
- FastEthernet 0/1 : 192.168.4.1
- Switch S2 :
- FastEthernet 0/1 : 192.168.3.0/27
- PC1 : 192.168.3.11
- PC2 : 192.168.3.12
- PC3 : 192.168.3.13
- PC4 : 192.168.3.14
- PC5 : 192.168.3.15

Configuring the devices :

Now that we have added our devices and connected them, we can start configuring them. We will start by configuring the router. Right-click on the router and select "CLI". This will open the command-line interface (CLI) for the router.

```
#enable
#configure terminal
#interface FastEthernet 0/0
#ip address {IP address} {subnet mask}
```

- It will connect you with wireless router.

As you can see in image below that system is connected. And PC1 card is active.

- Repeat same process on PC1 and PC2

Result:

Thus the configuration of wireless lan using Cisco packet tracer has been verified successfully.

Exno: 04
15/10/24

Configuration of wireless LAN using Cisco packet tracer.

AIM:

To design a topology with three PCs connected from linksys wireless router.

To perform following configuration:

- Configure static IP on PC and wireless router.
- Set SSID to mother network.
- Set IP address of router to 192.168.0.1, PC0 to 192.168.0.3, PC1 to 192.168.0.3 and PC2 to 192.168.0.4.
- Secure your network by configuring WAP key on router.
- Connect PC by using WAP key.

Step 1: Click on wireless router,

- Select Administration tab from top menu, set username and password to admin and click on save setting.
- Next click on wireless tab and set default SSID to mother network
- Now select wireless security and change security mode to WEP
- Set Key 1 to 0123456789.
- Again go in the end of page and click on save setting.
- Now it's time to connect PC's from wireless router, so do to click PC select desktop click on PC wireless.
- click on connect tab and click on refresh button
- It will ask for WAP key insert 0123456789 and click connect

Send a packet with the following:

- From computer 2
- To computer 5
- Packet type: "Ping request"

NMAP Host Discovery using ARP

When a privileged user tries to scan targets on a local network (Ethernet), Nmap uses ARP requests. A privileged user is root or a user who belongs to sudoers and can run sudo. When a privileged user tries to scan targets outside the local network, Nmap uses ICMP echo requests, TCP ACK to port 80, TCP, SYN to port 443, and ICMP timestamp request.

- From comp 1
- To comp 3
- Packet type: "ARP request".

TCP ACK Ping:

To utilize ACK ping in Nmap, which sends a packet with the ACK flag set, you need to run Nmap as a privileged user.

UDP Ping:

You can also employ UDP to check if the host is online. Unlike TCP SYN ping, sending a UDP packet to an open port typically doesn't elicit a response

- from computer 1
- To computer 1 (to indicate it is broadcast)
- Packet Type: "ARP Request"
- Data: computer 6 (because we are asking for computer 6 MAC address using ARP request)

- From computer 4.
- To computer 6 (to indicate it is broadcast)
- Packet Type: "ARP Request"
- Data: computer 6

Enumerating targets

Before delving into the details of scanning techniques, it's essential to define the targets we want to scan.

i) List
ii) Range
iii) Subnet

Discovering live hosts

TCP/IP layers:

- ARP from link layer.
- ICMP from network layer.
- TCP from transport layer.
- UDP from transport layer.

→ From computer 1
→ To computer 3
→ Packet type: "Ping Request".

NMAP to discover live hosts using Nmap scans (ARP, ICMP, TCP/UDP) on the tryhackme platform.

AIM: To discover live hosts using Nmap scan (ARP, ICMP, TCP/UDP) on the tryhackme platform room link: https://tryhackme.com/room/nmapcl.

Introduction:

When trageting a network, we need an efficient tool to handle repetitive tasks. This tool should help us find out which systems are active and what services are running on those systems. The tool that we will rely on is Nmap. The first question about finding live computers is answered in this room.

- Nmap live host discovery.
- Nmap basic port scan.
- Nmap advanced port scan.
- Nmap port port scan.

ARP scan: This scan uses ARP requests to discover live hosts.
ICMP scan: This scan uses ICMP requests to identify live hosts.
TCP/UDP ping scan: This scan sends packets to TCP ports and UDP ports to determine live hosts.

The figure displays two types of subnets:

/16 subnets: These have a subnet mask of 255.255.0.0 and can accommodate approximately 65 thousand hosts.

/24 subnets: These feature a subnet mask of 255.255.255.0 and can support around 250 hosts.

OUTPUT:

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION ===
Enter total number of frames to send : 8
Enter sender's window size : 4
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss) : 5
--------------------------------------
Sender window : [Frame 0 - Frame 3]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node A --> Sending Frame 3
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node B --> Received Frame 3, sending ACK 3
Node A --> ACK 3 received. sliding window forward.
--------------------------------------
Sender window : [Frame 4 - Frame 7]
Node A --> Sending Frame 4
Node A --> Sending Frame 5
    (Frame 5 LOST in transit!)
Node A --> Sending Frame 6)
Node A --> Sending Frame 7
Node B --> Frame 5 missing.
Node A --> Timeout!
--------------------------------------
Sender window : [Frame 4 - Frame 7]
Node A --> Sending Frame 5
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, ACK 5
Node B --> Received Frame 6, send ACK 6
Node B --> Received frame 7, ACK 7
```

```
def main():
    print("== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION
                                    (Go-back-N) ==\n")
    total_frames = int(input("Enter total number of frames to send  :"))
    window_size = int(input("Enter sender's window size  :"))
    if last_frame == -1:
        last_frame = None
    simulate(total_frames, window_size, last_frame)

if __name__ == "__main__":
    main()
```

RESULT :

Thus the above flow control at data link layer using sliding window protocol has been verified and executed successfully.

d. else (no loss in this batch):

i) The receiver accepts each frame in order and sends a cumulative acknowledgement (ACK) for it.

ii) on receiving the ACK for the last frame sent, the sender slides its window forward: base = next_seq.

5. Repeat step 4 until all N frames have been transmitted and acknowledged.

6. Stop the program.

CODE:

```
def simulate(total_frames, window_size, lost_frame):
    base = 0
    next_seq = 0
    while base < total_frames:
        window_end = min(base + window_size - 1, total_frames - 1)
        print("\n----------")
        print(f"Sender window: [Frame {base} - Frame {window_end}]")
        if lost_frame is not None and base <= lost_frame < next_seq:
            print(f"Node B --> Frame {lost_frame} missing. Discarding out-of-order frames after it.")
            print(f"Node A --> Timeout! Resending window starting from Frame {lost_frame}.")
            next_seq = lost_frame
            lost_frame = None
            continue
        for f in range(base, next_seq):
            print(f"Node B --> Received Frame {f}, sending ACK {f}")
        print(f"Node A --> ACK {next_seq-1} received. sliding window forward.")
        base = next_seq.
```

Flow Control at Data Link Layer Using Sliding Window Protocol.

AIM:

To write a program to implement flow control at the Data Link Layer using the sliding window protocol, and to simulate the flow of frames from one node to another.

ALGORITHM:

1. Start the program.
2. Read the total number of frames to be sent (N), the sender's window size (W), and (optionally) a frame number to simulate as lost during transmission.
3. Initialize base=0 and nextseq=0
4. Repeat while base <N:

a.) Determine the current sender window as the range [base, base+W-1].

b.) Send all frames with sequence numbers from nextSeq up to (base+W-1).

c) If the lost frame lies within the frames just sent:

i) The receiver discards the lost frame and any subsequent out-of-order frames.

ii) The sender times out and retransmits all frames starting from the lost frame (Go-back-N).

OUTPUT:

```
--- HAMMING CODE - ERROR DETECTION AND CORRECTION ---
Enter number of 4-bit data blocks in the stream : 2
--- Block 1 ---
Enter 4 data bits (space separated, e.g. 1011) : 1 0 1 1
Original Data Bits  : 1011
Encoded Hamming code (7 bits) : 0110011
Enter bit position to introduce error (1-7, 0 = no error) : 3
Transmitted code (with error induced at pos 3) : 0100011
Received code  : 0100011
checking for errors...
    Error detected at bit position : 3 -> correcting...
corrected code  : 0110011
Recovered data bits  : 1011

--- Block 2 ---
Enter 4 data bits (space separated) : 0 1 1 0
original data bits  : 0110
Encoded Hamming code (7 bits) : 1100110
Enter bit position to introduce error (1-7, 0 = no error) : 6
Transmitted code (with error induced at pos 6) : 1100100
Received code  : 1100100
checking for errors...
Error detected at bit position : 6 -> correcting...
Corrected code : 1100110
Recovered Data Bits : 0110
```

```
def to_str(bits):
    return "".join(str(b) for b in bits)

def main():
    print("--- HAMMING CODE - ERROR DETECTION AND CORRECTION ---\n")
    n = int(input("Enter number of 4-bit data blocks in the stream: "))
    for b in range(1, n+1):
        print(f"\n--- Block {b} ---")

    d = list(map(int, input("Enter 4 data bits (space separated): ").split()))
    h = encode(d)
    print(f"Original Data Bits : {to_str(d)}")
    print(f"Encoded Hamming code (7 bits): {to_str(h)}")
    pos = int(input("Enter bit position to introduce error (1-7, 0 = no error): "))
    if 1 <= pos <= 7:
        h[pos - 1] ^= 1
        print(f"Transmitted code (with error induced at pos {pos}): {to_str(h)}")
    else:
        print(f"Transmitted code (no error induced.): {to_str(h)}")

    print(f"Received code: {to_str(h)}")
    print("Checking for errors...")
    detect_and_correct(h)
    print(f"Corrected Code: {to_str(h)}")
    rd = [h[2], h[4], h[5], h[6]]
    print(f"Recovered Data bits: {to_str(rd)}")

if __name__ == "__main__":
    main()
```

Result:

The program to implement error detection and correction using the Hamming code concept was successfully written in python, executed and verified using a sample input data stream

6. Compute error position = c4 c2 c1

7. If Error position = 0, no error has occured.

8. Extract the original 4 data bits from the corrected 7-bit code word.

9. Repeat steps 3 to 8 for all n blocks in the data stream

10. Stop the program.

CODE:

```
def encode(d):
    h = [0] * 7
    h[2] = d[0]
    h[4] = d[1]
    h[5] = d[2]
    h[6] = d[3]

    h[0] = h[2] ^ h[4] ^ h[6]
    h[1] = h[2] ^ h[5] ^ h[6]
    h[3] = h[4] ^ h[5] ^ h[6]
    return h

def detect_and_correct(h):
    c1 = h[0] ^ h[2] ^ h[4] ^ h[6]
    c2 = h[1] ^ h[2] ^ h[5] ^ h[6]
    c4 = h[3] ^ h[4] ^ h[5] ^ h[6]
    error_pos = c1 * 1 + c2 * 2 + c4 * 4
    if error_pos != 0:
        print(f"Error detected at bit position: {error_pos} -> correcting...")
        h[error_pos - 1] ^= 1
    else:
        print("No error detected.")
    return error_pos
```

Exno: 6
5/8/26

Error Detection And Correction Using Hamming Code.

AIM:

To write a program to implement error detection and correction using the Hamming Code.

ALGORITHM:

1. Start the program.
2. Read the number of 4 bit data blocks (n) forming the input data stream.
3. For each 4-bit data block $D = d_1 d_2 d_3 d_4$, perform Hamming encoding:

a) Arrange 7 bits with parity bits at positions 1, 2, and 4, and data bits at positions 3, 5, 6 and 7.

b) Compute the parity bits using even parity:

$P_1 = d_1 \text{ XOR } d_2 \text{ XOR } d_4$ (covers bit positions 1, 3, 5, 7)

$P_2 = d_1 \text{ XOR } d_3 \text{ XOR } d_4$ (covers bit positions 2, 3, 6, 7)

$P_4 = d_2 \text{ XOR } d_3 \text{ XOR } d_4$ (covers bit positions 4, 5, 6, 7)

c) Form the 7-bit Hamming code word: $P_1 P_2 d_1 P_4 d_2 d_3 d_4$.

4. Simulate transmission by introducing a single-bit error to a chosen bit position in the code word.
5. At the receiver, recompute the check bits from the received code word:

$C_1 = P_1 \text{ XOR } d_1 \text{ XOR } d_2 \text{ XOR } d_4$

$C_2 = P_2 \text{ XOR } d_1 \text{ XOR } d_3 \text{ XOR } d_4$

$C_4 = P_4 \text{ XOR } d_2 \text{ XOR } d_3 \text{ XOR } d_4$

5. Create a filter to display only IP/ICMP packets and inspect the packets.

Procedure:

→ Select Local Area Connection in wireshark.
→ Go to capture option.
→ Select stop capture automatically after 100 packets.
→ Then click start capture.
→ Search ICMP/IP packets in search bar.

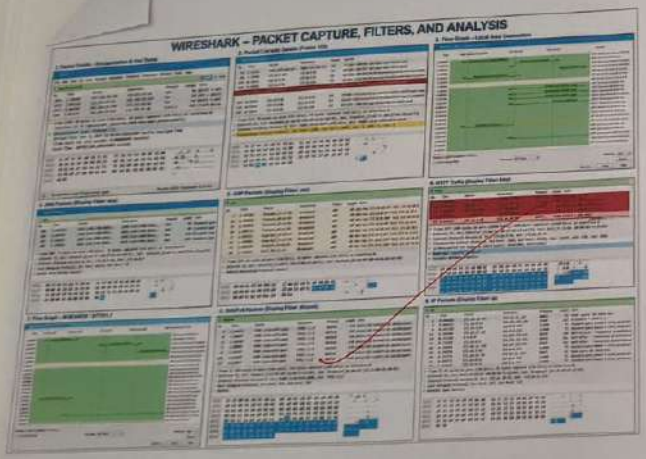
6. Create a filter to display only DHCP packets and inspect the packets.

Procedure:

→ Select Local Area connection in wireshark.
→ Go to capture option.
→ Select stop capture automatically after 100 packets.
→ Then click start capture.
→ Search DHCP packets in search bar.
→ Save the packets.

Result:

The experiments on packet capture tool wireshark has been executed successfully.



Inspecting packets:
click a packet to select it and you can dig down to view its details.

Capturing And Analysing Packets Using Wireshark Tool.

To filter, capture, views, packets in wireshark tool.

Capture 100 packets from the ethernet : IEEE 802.3 LAN interface and save it.

Procedure
- Select Local Area connection in wireshark
- Go to capture option
- Select stop capture automatically after 100 packets.
- Then click start capture.
- Save the packets.

OUTPUT

1. Create a filter to display only TCP/UDP packets, inspect the packets and provide the flow graph

Flow graph.

2. Create a filter to display only ARP packets and inspect the packets.

3. Create a filter to display only DNS packets and provide the flow graph.

4. Create a filter to display only HTTP packets and inspect the packets procedure.

Wireshark used for:

→ Network administrators: troubleshoot network problems.
→ Network security engineers: examine security problems.
→ Developers: debug protocol implementations
→ People: learn network protocol internals.

Getting wireshark:

wireshark can be downloaded for windows or macOS from its official website.

Capturing packets:

After downloading and installing wireshark, launch it and double-click the name of a network interface under capture to start capturing packets on that interface.

The "Packet List" Pane

The packet list pane displays all the packets in the current capture file.

The "Packet Details" Pane

The packet details pane shows the current packet in a more detailed form.

The "PacketBytes" Pane

The packet bytes pane shows the data of the current packet in a hexdump style.

Color coding

You'll probably see packets highlighted in a variety of different colors.

Filtering Packets:

If you're trying to inspect something specific, such as the traffic a program sends when phoning home, it helps to close down all other applications using the network so you can narrow down the traffic

EXNO:5
30/7/26

AIM:

Packet

Packet

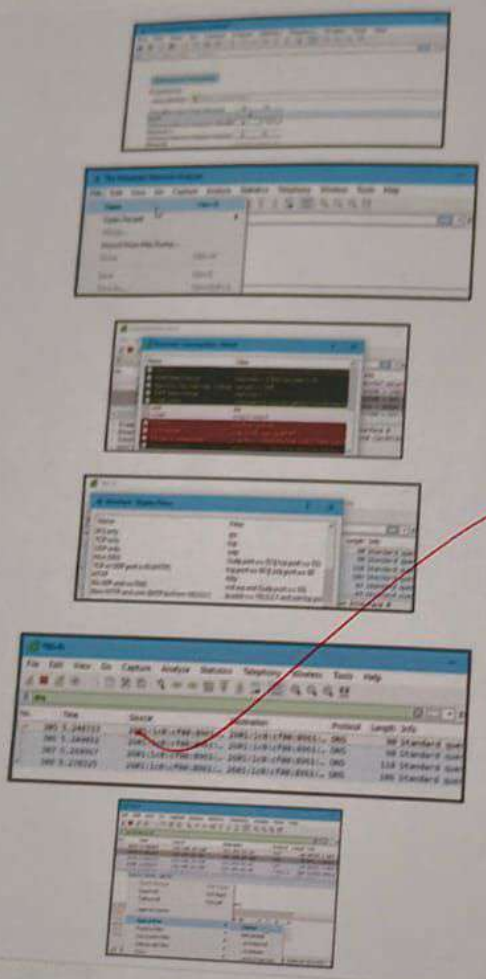
Descri

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packets.

What

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→ W



If you have a backup route, the backup route becoma the main route when you delete this main route.

For our example, use have a backup route and a main route for the host 30.0.0.100/8. The following Image shaw how to delete both routes.

Result:

Thus the simulation static routing Configuration using C[illegible] tra been implemented sucessfully

Verifying static routing:

On Router0, we configured two routes for network 30.0.0.0/8. These routes are Via Router1 and Via Router2. We set the first route as the main route and the second route as the backup route.

By sending ping requests to a PC of network 30.0.0.0/8 and tracing the path they take to reach the network 30.0.0.0/8. For this, you can use 'tracer' command on a PC of network 10.0.0.0/8. The 'tracer' command sends ping requests to the destination host and tracks the path they take to reach the destination.

We also configured a backup route for network 30.0.0.0/8. The router must put the backup route to the routing table and use it to forward data packets to network 30.0.0/8 when the main route fails.

To delete a static route, use the following steps.

- Use the 'show ip route static' command to print all static routes.
- Note down the route you want to delete.
- Use the 'no ip route' command to delete the route.

If you have a backup route, the backup route becomes the main route when you delete the main route.

Teacher's Sign / Remarks

Ex: 11(a) Simulate Static Routing Configuration Using Cisco Packet Tracer.
Date:

AIM:

Static routes are the routes you manually add to the router's routing table. The process of adding static routes to the routing table is known as static routing.

Creating, adding, verifying static routes.

Routers automatically learn their connected networks. We only need to add routes for the networks that are not available on the router's interfaces.

Router 1 requirements:

- Create two routes for network 10.0.0.0/8 and configure the first route as the main route and the second route as a backup route.
- Create two routes for network 40.0.0.0/8 and configure the first route as the main route and the second route as a backup route.
- Verify the router adds only main routes to the routing table

Router 2 requirements

Create static routes for network 10.0.0.0/8 and network 30.0.0.0/8 and verify the router adds both routes to the routing table.

step: 2 configure the laptop
step: 3 configure the PC
step: 4 configure the internet cloud
step: 5 configure the cisco.com server

Result:

Thus, the design and configuration of an internetwork using wireless router, Http server and internet cloud.

Aim: 10(b) Design and configure an internetwork using wireless router, DHCP server and internet cloud.

Aim:

To design and configure an internetwork using wireless router, DHCP server and internet cloud.

PART 1: Build a simple network in the logical topology workspace

PART 2: Configure the network devices.

PART 3: Test connectivity between network devices

PART 4: Save the file and close packet tracer.

Step: 1 Launch Packet Tracer.

Step: 2 Build the topology

a) Add network devices to the workspace.

Using the device selection box, add the network devices to the workspace as shown in the topology diagram.

b) Change display names of the network devices.

. Type the new name of the device into the display name box as show in the figure below.

c) Add the physical cabling between devices on the workspace

Using the device selection box, add the physical cabling between devices on the workspace as shown in the topdesy digram.

PART 2 . configure the network devices.

Step: 1 configure the wireless router.

a) create the wireless network on the wireless router.

b) click on the GUI setting tab.

Aim:

Web servers create log files containing information such as

client IP
Date and time
RequestedURL
HTTP method
HTTP statuscode
Bytes transferred
Referrer
User agent

Example input:

access.log

OUTPUT:

webalizer generates an HTML-based statistics containing

webalizer statistics.

Total requests : 1250
Total visits : 480
Total pages : 920
Total bytes : 154 MB
Successful requests : 1120
404 Errors : 45

Ex15 Analyse various types of servers using webalizer tool.

AIM:

To analyze web server log files using the webalizer tool and study website traffic, visitor request, pageview, bandwidth usage and other server statistics.

Software Required:

* operating system: ubuntu/linux.
* Apache web server.
* webalizer tool
* Terminal.

Types of server Analysed.

- Webserver : Handles HTTP/HTTPS request from browsers.
- FTP server : Transfer files between computer.
- Mail server : Sends and receives emails.
- DNS server : Converts domain name into IP addr
- proxy server : Forward request between clients other server.

PROGRAM:

```
from scapy.all import sniff, IP
def packet_callback(packet):
    if IP in packet:
print("protocol:", packet[IP].proto)
print("source IP:", packet[IP].src)
print("destination IP:", packet[IP].dst)
print("-" * 40)
print("Packet sniffing started...")
print("Capturing 10 packets...")
sniff(prn=packet_callback, count=10)
print("Packet Sniffing completed.")
```

OUTPUT:

```
Protocol: 17
SourceIP: 172.16.71.214
DestinationIP: 239.255.255.250
```

AIM:

To write

raw sockets to

basic packet

destination IP

ALGORITHM:

1. Start the
2. Initialize the
3. Create a raw
4. Bind the
5. Enable
6. Wait for
7. Read the
8. Extract the
9. Extract the
10. Identify the
11. Display the
12. Continue
is stopped
13. Close the

RESULT:

Thus,
was implemented
network packets
destination

Ex. 14.
8/11/25

Write a code using RAW sockets to implement packet sniffing

AIM:

To write and implement a C program using raw sockets to sniff network packets and display basic packet information such as source IP address, destination IP address, protocol, and packet size.

ALGORITHM:

1. Start the program.
2. Initialize the socket library.
3. Create a raw socket using the IP protocol.
4. Bind the socket to the local network interface.
5. Enable packet reception.
6. Wait for incoming network packets using recvfrom().
7. Read the IP header from the captured packet.
8. Extract the source IP address.
9. Extract the destination IP address.
10. Identify the protocol used by the packet.
11. Display the packet size and extracted information.
12. Continue capturing packets until the program is stopped.
13. Close the socket and terminate the program.

RESULT:

Thus, a packet sniffer using raw sockets was implemented successfully. The program captures network packets and displays their source IP address, destination IP address, protocol and packet size.

Ex:13
24/9/25

Implement your own ping program.

AIM:

To implement a simple Ping program using ICMP packets to check whether a destination host is reachable and measure the response time.

ALGORITHM:

1. Start the program.
2. Initialize the socket library.
3. Create a raw socket using ICMP protocol.
4. Specify the destination IP address or hostname.
5. Create an ICMP echo request packet.
6. Record the start time.
7. Send the ICMP packet to the destination.
8. Wait for an ICMP Echo reply.
9. Record the end time.
10. Calculate the round Trip Time (RTT)
11. Display the destination, packet size, and response time.
12. Repeat the process for the required no of packets.
13. Close the socket and terminate the program.

RESULT:

Thus, the own ping program was implemented successfully using ICMP packets. It sends Echo request packets to the destination host, receives echo reply packets, and calculates the round-trip time to determine network connectivity.

C.) UDP chat server.

1.) Start the server.
2.) Create a UDP socket.
3.) Bind the socket to a port number.
4.) Receive a message from the client.
5.) Display the message.
6.) Enter and send a reply to the client.
7.) Repeat until "exit".
8.) Close the socket.

D.) UDP client chat

1.) Start the client.
2.) Create a UDP socket.
3.) Enter a message
4.) Send the message to the server.
5.) Receive the server's reply.
6.) Display the reply.
7.) Repeat until "exit".
8.) Close the socket.

RESULT:

The chat client-server application was successfully implemented using TCP and UDP sockets, and messages were successfully exchanged between the client and server.

Implement chat client server using TCP/UDP sockets.

AIM:

To implement a chat client server application using TCP and UDP sockets, allowing the client and server to exchange messages.

ALGORITHM:

A.) TCP chat server

1. Start the server.
2. Create a TCP socket.
3. Bind the socket to an IP address and port number.
4. Wait for the client to connect.
5. Accept the client connection.
6. Receive message from the client
7. Send a reply to the client.
8. Repeat the communication until "exit".
9. close the connection.

B.) TCP chat client

1.) Start the client.
2.) Create a TCP socket.
3.) Connect to the server.
4.) Enter a message.
5.) Send the message to the server.
6.) Receive the server's reply.
7.) Display the reply.
8.) Repeat until "exit"
9.) Close the connection.

a) UDP Echo server:

1) Start the server.
2) Create a UDP socket.
3) Receive a message from the client.
4) Read the message.
5) Send the same message back to the client.
6) Repeat until "exit".
7) Close the socket.

b) UDP echo client

1) Start the client.
2) Create a UDP socket.
3) Enter a message.
4) Send the message to the server.
5) Receive the echoed message.
6) Display the message.
7) Repeat until "exit".
8) Close the socket.

RESULT: The echo client-server application was successfully implemented using TCP and UDP sockets. The client sends a message to the server, and the server successfully echoes the same message back to the client.

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Ex: 12(a)
17/9/26

Implement echo client server using TCP/UDP sockets.

AIM:

To implement an echo client-server application using TCP and UDP sockets, where the client sends a message to the server and the server echoes the same message back to the client.

ALGORITHM:

A) TCP Echo server

1. Start the TCP server
2. Create a server socket with a specified port number.
3. Wait for a client connection using accept().
4. Create input and output streams.
5. Receive the message from the client.
6. Send the message back to the client.
7. Continue until the client sends 'exit'.
8. Close the socket and server.

B) TCP echo client

a.) Start the client
b.) Connect to the server using IP address and port number.
c.) Enter a message
d.) Send the message to the server.
e) Receive the echoed message
f) Display the message.
g) Repeat until "exit".
h.) Close the connection.

Router 2

Now router have information about the router will not exchange this information implement RIP routing protocol.

- Enable RIP routing protocol from global
- Tell RIP routing protocol which need advertise.

Router 0

router rip command tell router to enable the RIP routing protocol. network command allow us to specify the networks which we want to advertise. we only need to specify the networks which are directly connected with the router.

Router 1

Router 2

Router 2 (config)# router rip

RIP protocol automatically manage all router for us. If one route goes down, it automatically switches to another available.

currently there are three routes between PC0 and PC1

Result:

Thus the simulation RIP using CISCO packet tracer has been executed successfully.

Ex: 11(b) Simulate RIP Using Cisco Packet Tracer.

AIM:

Assign IP address to PCs: double click PCs and click desktop menu item and click IP configuration. Assign IP address referring the above table.

Assign IP address to interfaces of routers.
Double click router0 and click CLI and press enter key to access the command prompt of Router0.

We need to configure IP address and other parameters on interfaces before we could actually use them for routing. Interface mode is used to assign IP address and other parameters.

From global configuration mode we can enter in interface mode. From there we can configure the interface.

interface fastEthernet 0/0 command is used to enter in interface mode.
ip address 10.0.0.1 255.0.0.0 command will assign IP address to interface.

no shutdown command will bring the interface up.
exit command is used to return in global configuration mode.

Serial interface needs two additional parameters clock rate and bandwidth.